BIG DATA PROCESSING · GROUP ASSIGNMENT 2 · TEAM BIG

Pricing Airbnb listings in Switzerland

and what happened when a third client, the canton of Vaud, signed up a week later

A · Geneva B · Zurich C · Vaud

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THE TASK

Two cities to learn from, one canton to prove it on

A

Geneva

Training city

2,538 listings · 23 Sep 2026

B

Zurich

Training city

3,174 listings · 25 Sep 2026

C

Vaud

New client, kept unseen until week 2

5,235 listings · 5 Oct 2026

Vaud is a **canton**, not a city: Lausanne, lake towns and Alpine ski resorts. We chose it knowing the transfer would be harder.

EXPLORATORY ANALYSIS

A third of listings have no price: they are inactive, not missing

Geneva



70% active · 30% no price

Zurich



66% active · 34% no price

Vaud



69% active · 31% no price

98% vs 6–10%

share without a price among listings closed all year vs open ones. We train only on priced listings and never impute the target.

Outliers by price per guest

13 implausible prices (a room at CHF 11,382) excluded; 26 expensive but real villas kept.

LEAKAGE REVIEW

Two columns are the price in disguise

IS THE PRICE

price_quote_price_per_night

identical to price in 100% of listings

COMPUTED FROM IT

estimated_revenue_l365d

= price × occupancy, a 100% match

Also excluded: outcomes of the price

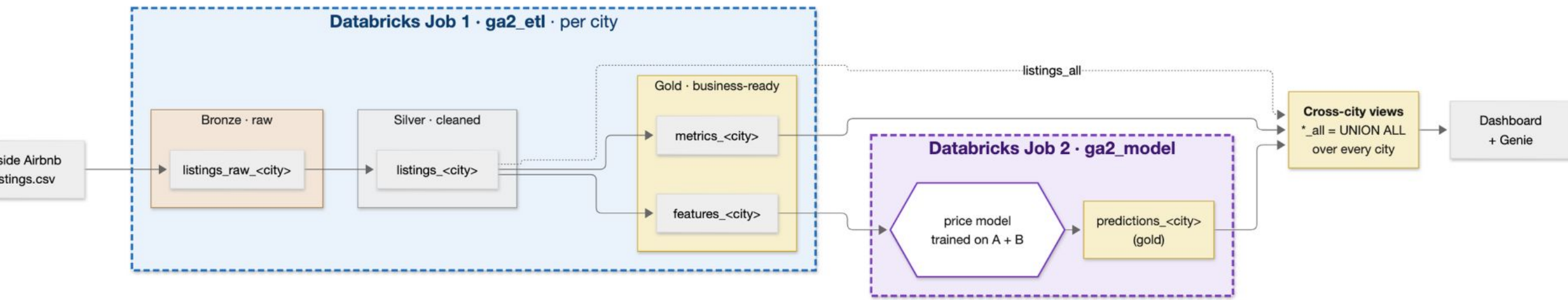
availability, occupancy, recent reviews: how much gets booked at that price

Allowed: what a host decides

room type, guests, bedrooms, bathrooms, location, minimum stay, review history

Solution architecture

Two Databricks Jobs: **ga2_etl** (run per city: A Geneva · B Zurich · C Vaud), then **ga2_model** (train on A + B, predict)



GENEVA AGAINST ZURICH

Zurich is 20% pricier overall

Median price per night

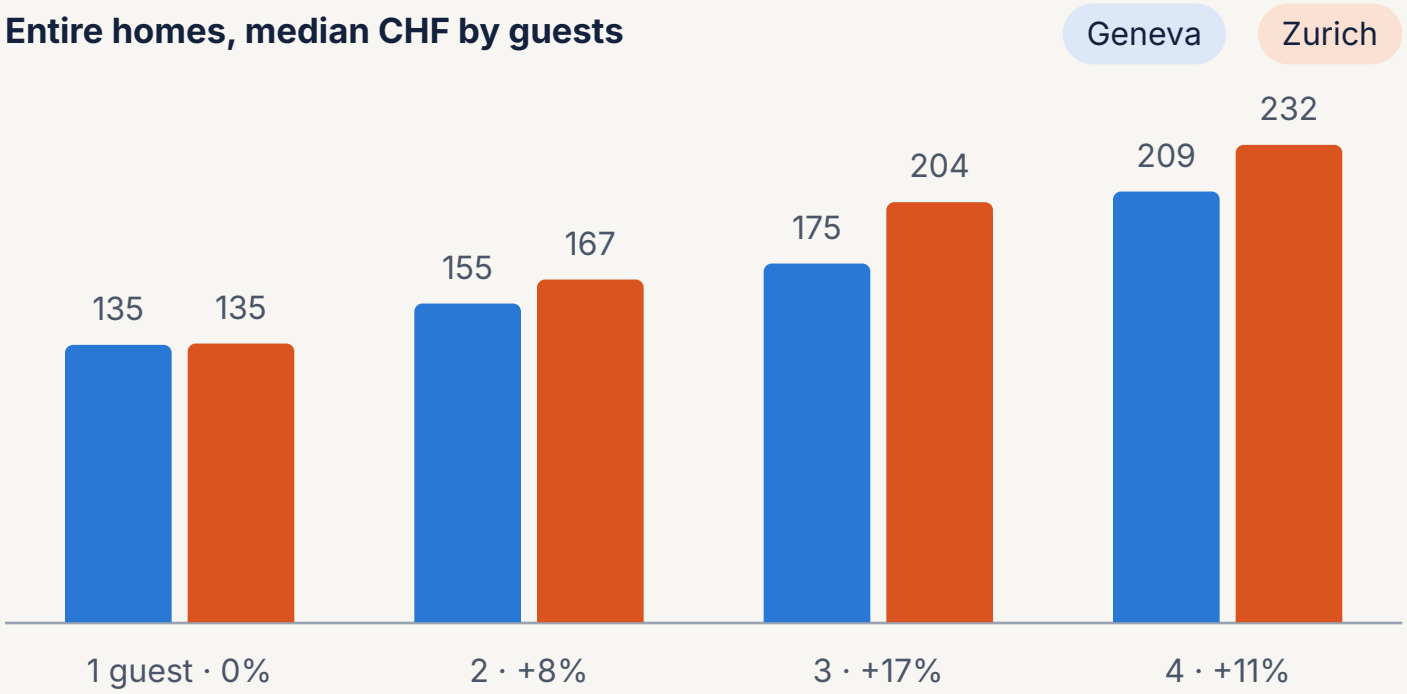
154 → 184

CHF, Geneva → Zurich: **+20%**

Listings for 4+ guests

26% → 34%

bigger flats lift Zurich's median



PRICE MODEL

Boosted trees, constrained by rules a host would agree with

↑ Can only raise the price

guests · bedrooms · bathrooms · amenities

↓ Can only lower it

distance to the centre · shared bathroom · minimum stay

◆ Free to learn

room type · city · reviews and rating · superhost · host's listing count

Target
log price

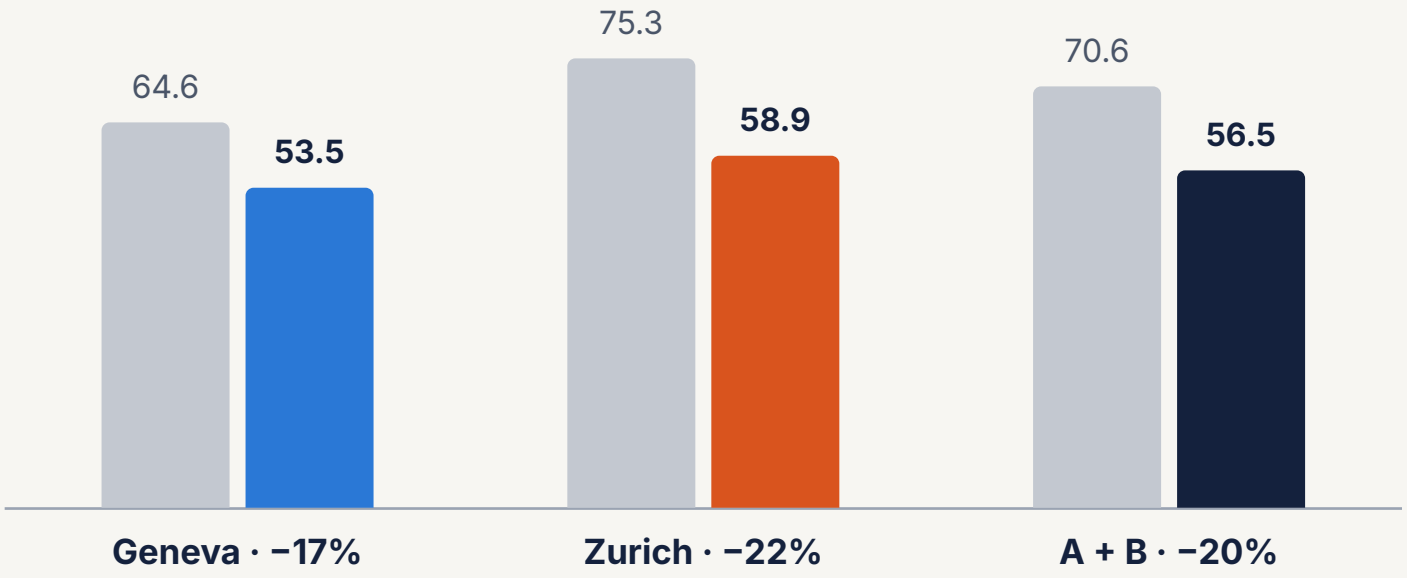
Loss
absolute error

Model choice
5-fold CV on train rows

Test set
20% per city, used once

PRICE MODEL · TEST SET

The model’s error is 20% below the baseline on Geneva and Zurich



baseline (left bar)

model (right bar)

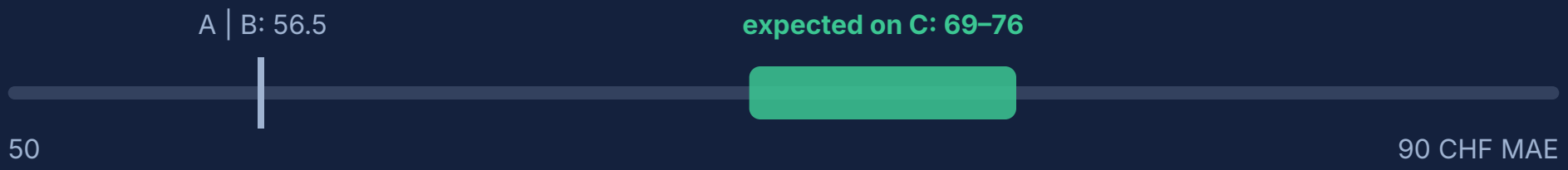
Baseline: the median training price for the same room type and number of guests.

MAE in CHF per night on 800 held-out listings.

CITY C · BEFORE OPENING ITS DATA

We predicted Vaud's error before we looked at it: 69–76 CHF

Rehearsal with what we had: train on Geneva and predict Zurich, then the reverse. An unseen city cost 22–35% more error than A | B.



CITY C · FIRST RUN, CODE UNCHANGED

Nothing broke on the first run, but the errors were systematic

5 of 5
pipeline tasks ran, no failures

5,235
listings, every one priced

71.9
CHF MAE on test, inside 69–76

Lausanne over-priced by 21%

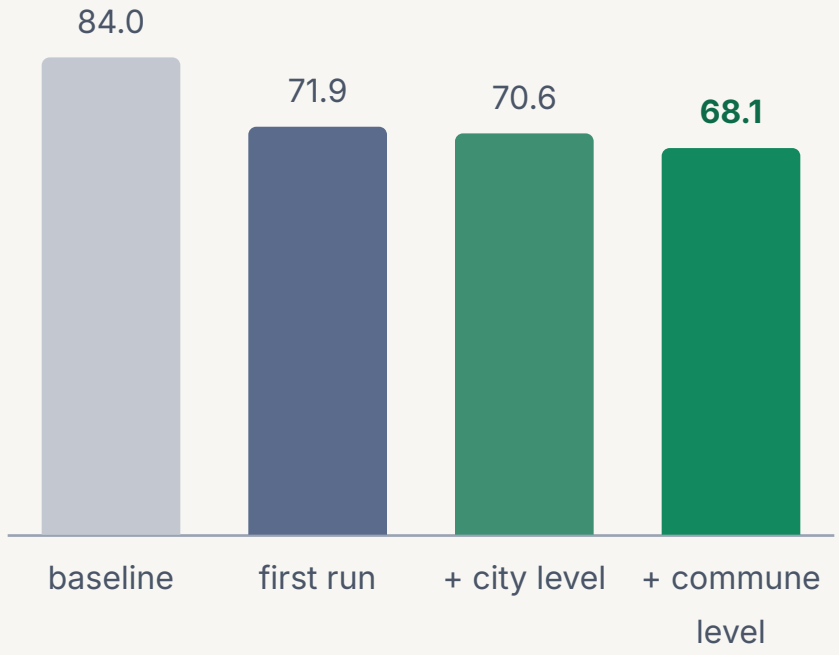
Vaud is a city label the model never saw, so it got a training city's price level. Lausanne (median CHF 131) is cheaper than central Geneva or Zurich.

43% of listings 30+ km out

The farthest listing in A or B is 16 km from the centre. In Vaud that distance means Alpine chalets: MAE ~100 CHF there vs ~50 in Lausanne.

CITY C · WHAT WE CHANGED

Two model changes took Vaud from 71.9 to 68.1, the pipeline needed none



Drop the city feature
one city-free model plus a price level learnt from Vaud's known prices

MODEL
70.6

Add a price level per commune
one level per Vaud municipality, shrunk to zero where few prices are known

MODEL
68.1

No pipeline fix needed for Vaud
two bugs were found on the first Databricks run of A and B

PIPELINE
0

CITY C · WHERE THE BIAS WENT

Local price levels removed the bias in Lausanne, Montreux and Leysin



CITY C · THE THREE NUMBERS

Vaud is harder for any method, and the model still keeps its edge

Baseline on C	Model on A + B	Model on C, first run	Model on C, after changes
84.0	56.5	71.9	68.1

Harder for everyone

The baseline itself is 19% worse on Vaud (84.0 vs 70.6): chalets spread prices more than city flats.

The model transfers

It beats the baseline by 20% on A + B, by 14% on Vaud's first run and by 19% after the changes.

What is left: the Alps

No altitude, season or ski-resort signal: a chalet 30 km out looks like a big suburban house.

DEMO · AI/BI DASHBOARD IN DATABRICKS

Our dashboard has three pages, and each page title states its main finding

PAGE 1

City overview

KPIs per city, room-type mix, price bands, top neighbourhoods

Finding in the page title

Entire homes are 70–76% of listings in every city

PAGE 2

Geneva vs Zurich

side-by-side metrics, like-for-like prices, listing sizes

Finding in the page title

Zurich is 20% pricier, but only 8 to 11% for a same-size flat

PAGE 3

Price model

model vs baseline, predicted vs actual, worst listings, Vaud before and after

Finding in the page title

The largest errors are in the Vaud Alps

Same colours everywhere: **Geneva**, **Zurich**, **Vaud**, grey = baseline. Every model number is on the test set.

DEMO · GENIE AGENT SETUP

Five base rules in Genie's instructions, then a test on 11 questions

BASE RULES

1. **City letters:** A = Geneva, B = Zurich, C = Vaud; prices in CHF per night
2. **City totals** only from the "All" row, never by adding up room types
3. **Prices** only from active, plausible listings; "median" is the 50th percentile
4. **Model accuracy** only on test listings; an error below 0 means under-priced
5. **Model vs baseline** from the evaluation tables, Vaud with both runs

HOW WE TESTED

4 questions from the assignment

4 / 4

correct, each with an example query

7 new questions, no example query

4 / 7 → 7 / 7

3 wrong at first, each fixed by a rule

DEMO · GENIE AGENT

We tested Genie on new questions, and each wrong answer became a rule

Question	Wrong answer	Rule we added	Right answer
Most common room type in Vaud?	“All”, 100%	Room-type questions skip the “All” row	**Entire home, 72.6%**
The model’s error on city C?	68.1, nothing else	Always give both runs and the baseline	**71.9 → 68.1, baseline 84.0**
Most expensive municipalities in C?	Commugny: 6 listings, CHF 1,246	Rank only places with 10+ known prices	**Rougemont, CHF 367**

CHALLENGES

Our hardest problems came from the data, not from the pipeline

One centre for a whole canton

“Distance to the centre” means a suburb in Geneva but a ski resort in Vaud.

What we did: learnt a price level per municipality from the client’s own known prices.

Data we do not have

No season, altitude or fees, and no price at all for inactive listings.

What we did: stated them as known limits instead of guessing proxies.

TEAM

Who did what

Pavlo Ruzhylo

DATA AND PIPELINE

- repository and data loading
- exploratory analysis and leakage review
- architecture
- bronze, silver and gold pipeline, ETL job
- first price model

Oleksii Lasiichuk

MODEL AND CITY C

- model choice, monotonic boosting
- prediction job, transfer rehearsal
- city C first run, changes, report
- pipeline fixes on Databricks

Ivan Maksymchuk

DASHBOARD AND GENIE

- AI/BI dashboard, three pages
- Genie Agent, rules and tests
- presentation and README

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Thank you for your attention

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